

PRESANNAKUMAR S

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Software Developer | Ai & Machine Learning Enthusiast | Full-Stack Developer

AI-focused Electronics and Communication Engineering student (CGPA: 8.97/10) with proven expertise in Java, Machine learning and full-stack web development. Experienced in building Web development, AI/NLP-powered systems, quantum-inspired security platforms, and IoT-integrated solutions. Seeking software engineering roles at product and service companies with a focus on scalable, data-driven, and optimized system development.

EDUCATION

B.E. Electronics and Communication Engineering	2023 – 2027
<i>Sathyabama Institute of Science and Technology, Chennai CGPA: 8.97 / 10.0</i>	

TECHNICAL SKILLS

Programming Languages	Java (Primary), Python, C, SQL, HTML, CSS
Tools & Technologies	MySQL, Power BI, OpenCV, VS Code, Git, Web Development, MATLAB, Cadence Virtuoso, Computer Networks, IOT
Technical Skills	Machine Learning, Full-Stack Development, Database Management Systems, Data Structures and Algorithms, Cloud Computing
Soft Skills	Problem Solving, Leadership, Negotiation, Communication, Teamwork, Critical Reasoning

PROJECTS

Digital Twin Enabled Cognitive Edge IoT Framework	2026
<i>ESP32-S3 TinyML Next.js real-time Dashboard</i>	
- Developed a TinyML - based emergency vehicle detection and traffic signal prioritization system. - Built a 3D Digital Twin dashboard for real-time traffic monitoring. - Integrated IoT communication for low-latency traffic control.	
AI Resume Ranking System	2026
<i>Python NLP Scikit-learn TF-IDF</i>	
- Engineered an AI-powered resume screening system using NLP and TF-IDF vectorization to rank resumes against job descriptions, improving candidate shortlisting efficiency by automating manual scoring. - Implemented keyword extraction and semantic matching algorithms to quantify resume-job alignment with a ranked scoring output. - Designed a scalable pipeline capable of processing multiple resumes in batch, reducing recruitment turnaround time.	
Quantum-Inspired Secure Communication Platform	2026
<i>Python BB84 Protocol XOR Encryption Real-Time Simulation</i>	
- Developed a quantum-inspired encryption platform integrating QRNG-based key generation and BB84 protocol simulation with multi-layer XOR encryption to achieve high data security and diffusion. - Built a real-time attack simulation engine supporting intercept-resend and brute-force scenarios with live QBER (Quantum Bit Error Rate) threshold-based intrusion detection. - Created an interactive visualization dashboard demonstrating quantum key distribution flow, encryption stages, and attack impact dynamically.	

Business Websites (3 Projects)	2025–2026
<i>HTML CSS JavaScript Responsive UI SEO</i>	
- Developed full-stack responsive websites for a Construction Company, and a Two Hospitals, each featuring service listings, gallery, contact/enquiry system, and booking functionality. - Implemented SEO optimization, performance tuning, and complex animated UI patterns for the Hospital website, improving user engagement and page load speed. - Deployed all projects with domain configuration and hosting, delivering production-grade web solutions.	

INTERNSHIPS

BSNL Chennai — Network & Telecom Intern	2024
Gained hands-on exposure to data networking infrastructure, telecom protocols, and network topology management in a real-world enterprise environment.	
Eaton Power Quality Pvt Ltd — Embedded Systems Intern	2025
Worked with embedded control systems in an industrial power quality environment; analysed circuit behaviour and system diagnostics under supervision.	

ACHIEVEMENTS & CERTIFICATIONS

- NPTEL Certified — IoT Fundamentals and Sensor Technologies - CGPA: 8.97/10.0 — Consistent academic excellence across all semesters - Developed 20+ projects spanning AI, quantum cryptography, IoT, and full-stack web development - Proficient in Tamil, English, Kannada, and French	
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